Student Learning Management and Collaboration System

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Spring 2026

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Executive Summary

The **Student Learning Management and Collaboration System (EduCollab)** is an online academic platform designed to provide students with a structured environment for learning, communication, and collaboration. Unlike general social media applications that often cause distractions, this system focuses entirely on educational activities by enabling users to manage academic tasks, monitor progress, and interact with peers in a productive manner.

The proposed system integrates several important features, including marks tracking, goal setting and progress monitoring, study group collaboration, resource sharing, messaging, notifications, and reputation-based leaderboards. These functionalities encourage active participation, improve organization, and motivate students to achieve their academic objectives. Teachers and mentors can also monitor student engagement and provide timely feedback and guidance.

Overall, EduCollab offers a centralized and user-friendly solution that enhances digital learning experiences. By combining academic management with collaborative tools and gamification elements, the system promotes effective communication, continuous improvement, and increased student productivity. Its scalable and accessible design makes it suitable for educational institutions seeking to support modern teaching and learning practices.

Contents

Table of tables:

1.Introduction

The Student Learning Management and Collaboration System (EduCollab) is an online platform designed to support students in managing their academic activities through collaboration, progress tracking, and resource sharing. It provides a centralized and distraction-free environment that enhances communication, productivity, and overall learning experiences.

1.1 Problem Statement

In the modern educational environment, students rely on digital platforms for communication and collaboration. However, commonly used apps like WhatsApp and Facebook are not designed for academic purposes, leading to distractions and unorganized communication.

Within a **Student Learning Management and Collaboration System**, there is a lack of a structured platform where students can manage their learning effectively.

Therefore, there is a need for a **Student Learning Management and Collaboration System** that provides structured learning, collaboration, progress tracking, and improved academic productivity in one platform.

1.2 Proposed Solution

To overcome these challenges, a **Student Learning Management and Collaboration System** is proposed. This system aims to provide a structured and distraction-free environment tailored specifically for academic collaboration

The system will allow students to create or join study groups based on subjects or courses, enabling focused and organized discussions.

One of the core features of the system is the **reputation mechanism**, where students earn points or ratings based on their contributions.This system will encourage healthy competition, improve engagement, and ensure fair recognition of efforts.

1.3 Target Users

1. Students:

Students are the primary users of the system. They will use the platform to collaborate with peers, join or create study groups, set academic goals, and track their progress

2. Group Admin / Leader:

Group admins or leaders are responsible for managing study groups. They can create groups based on specific subjects.

3. Teachers / Mentors:

Teachers or mentors act as supervisors within the system. They can monitor student performance, review progress reports, and provide feedback or guidance when necessary.

2.Business Scope & Feasibility

2.1 Business Goals

The main goal of the proposed Student learning Management and system is to improve academic collaboration and productivity among students.

i. To provide a centralized platform for student collaboration
ii. To enhance productivity by reducing distractions
iii. To enable students to set and track academic goals

2.2 Stakeholders

Stakeholders are individuals or groups who have an interest in the system or are affected by it.

1. Students:

They are the primary users of the system. Their main interest is to collaborate efficiently, improve academic performance, and track their goals.

2. Teachers / Mentors:

They use the system to monitor student progress, provide feedback, and guide students. Their goal is to ensure effective learning outcomes.

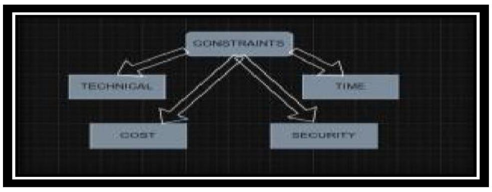
3. Development Team:

The team responsible for designing, developing, and maintaining the application. Their interest is to build a reliable and user-friendly system.

4. Educational Institutions:

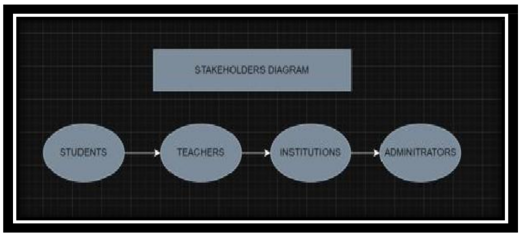
Schools or universities may adopt this system to enhance digital learning and collaboration among students.

2.3 Constraints



Every project has certain limitations that must be considered during development.

1. Technical Constraints: The system requires stable internet connectivity and compatible devices such as smartphones or computers. Limited technical resources may affect system performance and scalability.



2. Time Constraints: The project must be completed within a limited timeframe. This may restrict the number of features that can be implemented.

3. Budget Constraints: Limited financial resources may affect the choice of tools, technologies, and hosting services.

4. User Adoption Constraints: Students and teachers may take time to adapt to a new system, especially if they are used to existing platforms.

5. Security & Privacy Constraints: The system must ensure protection of user data, which requires proper security measures and compliance with basic data protection standards.

3.Tools & Technologies

Development Stack

The development of the Student Collaboration Application will use modern web technologies to ensure performance, scalability, and user-friendly design.

Frontend (User Interface): The frontend will be developed using React. It allows the creation of interactive and responsive user interfaces. React helps in building reusable components, making the application faster and easier to maintain.

Backend (Server-side):

The backend will be developed using Node.js with Express.js. This combination enables efficient handling of requests, user authentication, and communication between the frontend and database.

Database:

The system will use MongoDB to store user data, goals, group information, and reputation scores. MongoDB is flexible and suitable for handling.

4.Work Breakdown Structure

- ## Introduction to WBS

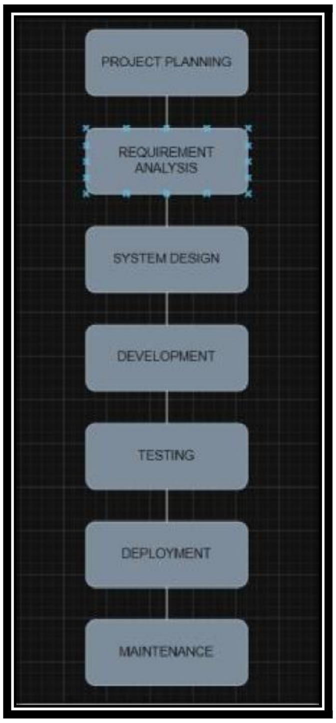
For the Student Collaboration Application, the project is divided into different phases such as planning, design, development, testing, and deployment.

- ## Task Breakdown

The following is the hierarchical breakdown of tasks involved in the project:

4.1.Project Planning

- Define project objectives
- Identify system requirements
- Assign roles to team members
- Select tools and technologies



4.2.Requirement Analysis

- Gather requirements from users (students/teachers)
- Analyse functional and non-functional requirements
- Prepare requirement specification document

4.3.System Design

- Design system architecture
- Create UI/UX design (screens, layout)
- Design database structure
- Prepare system flow diagrams

4.4.Development

- Frontend development (user interface)
- Backend development (server and APIs)
- Database implementation
- Integration of all components

4.5.Testing

- Unit testing (individual components)
- Integration testing

- System testing
- Fixing bugs and errors

4.6.Deployment

- Prepare final version of the system
- Deploy application on server
- Ensure system is accessible to users

4.7.Maintenance

- Monitor system performance
- Fix issues after deployment
- Update features if required

5.Project Timeline

- ### 8-Week Timeline Plan

The project will be completed according to the following schedule:

5.1.Week 1: Project Planning

- Define project objectives
- Identify team roles
- Select tools and technologies

5.2.Week 2: Requirement Gathering & Analysis

- Collect requirements from students and teachers
- Analyse functional and non-functional requirements
- Prepare requirement document

5.3.Week 3: System Design (Part 1)

- Design system architecture
- Create basic UI/UX layouts

5.4.Week 4: System Design (Part 2)

- Finalize UI design
- Design database structure
- Prepare system diagrams

5.5.Week 5: Development (Frontend)

- Develop user interface using React
- Create main pages (login, dashboard, groups)

5.6.Week 6: Development (Backend)

- Develop server using Node.js
- Implement APIs and database integration

5.7.Week 7: Testing

- Perform unit and integration testing
- Identify and fix bugs
- Improve system performance

5.8.Week 8: Deployment & Finalization

- Deploy the application
- Prepare final documentation
- Make final improvements

- **Gantt Chart**

Table 1 gantt chart

TASK	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
planning	✓							
requirement		✓						
design			✓	✓				
development					✓	✓		
testing							✓	
deployment								✓

6.Functional Requirements (FRs)

These are the following functional requirements for the Student Collaborative Application.

6.1. FR1: User Registration

The system shall allow new users (students, teachers) to create an account by providing basic information such as name, email, and password.

6.2. FR2: User Login & Authentication

The system shall allow registered users to securely log in using their credentials and access their accounts.

6.3. FR3: Profile Management

The system shall allow users to create and update their profiles, including profile picture, bio, and academic interests.

6.4. FR4: Create Study Groups

The system shall allow users (especially group admins) to create study groups based on subjects or courses.

6.5. FR5: Join Study Groups

The system shall allow students to search and join existing study groups

6.6. FR6: Group Management

The system shall allow group admins to add/remove members, assign roles, and manage group activities.

6.7. FR7: Goal Setting

The system shall allow users to set personal and group academic goals (e.g., completing assignments, exam preparation).

6.8. FR8: Goal Tracking

The system shall track users' progress toward their goals and display progress updates.

6.9. FR9: Resource Sharing

The system shall allow users to upload and share study materials such as notes, documents, and links.

6.10. FR10: Messaging & Discussion

The system shall provide a communication feature for group discussions and messaging between users

6.11. FR11: Reputation System

The system shall assign reputation points to users based on their contributions, such as sharing resources, helping others, and completing tasks.

6.12. FR12: Leaderboard

The system shall display a leaderboard showing top-performing users based on reputation points.

6.13. FR13: Notifications

The system shall send notifications for important events such as new messages, goal deadlines, or group updates.

6.14. FR14: Teacher/Mentor Monitoring

The system shall allow teachers or mentors to monitor student activities, track progress, and provide feedback.

6.15. FR15: Search Functionality

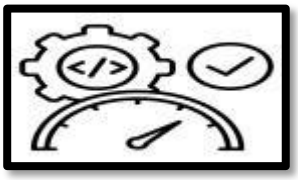
The system shall allow users to search for study groups, users, or shared resources easily.

7.Non-Functional Requirements (NFRs)

These are the following non-functional requirements for the Student Collaborative Application

7.1. NFR1: Performance

The system should respond quickly to user actions. Page loading time should not exceed 2–3 seconds under normal conditions to ensure a smooth user experience.



7.2. NFR2: Usability

The application should have a simple, intuitive, and user-friendly interface so that students and teachers can easily navigate and use all features without difficulty.

7.3. NFR3: Security

The system must ensure secure user authentication and protect user data (such as passwords and personal information) using encryption and secure login mechanisms.

7.4. NFR4: Availability

The system should be available to users 24/7 with minimal downtime, ensuring students can access it anytime for study and collaboration.

7.5. NFR5: Scalability

The system should be able to handle an increasing number of users, groups, and data without performance issues as usage grows.

7.6. NFR6: Reliability

The application should function correctly without crashes or errors. It should consistently perform its intended functions even under heavy usage.

7.7. NFR7: Maintainability

The system should be easy to update, modify, and maintain. Developers should be able to fix bugs or add new features without affecting existing functionality

7.8. NFR8: Compatibility

The application should work properly on different devices such as smartphones, tablets, and computers, and support multiple browsers.

7.9. NFR9: Data Backup & Recovery

The system should regularly back up data and provide recovery mechanisms in case of system failure or data loss.

8.Requirement Engineering Process

Requirement Engineering is a disciplined process used to discover, analyse, document, validate, and manage the requirements of a software system. For the Student Collaborative System with Goal Tracking and Reputation System, this process ensures that all user needs are properly understood and converted into clear system specifications. The main activities are described below:

8.1. Elicitation

Requirement elicitation is the first and most important phase of the Requirement Engineering Process.

- Assigning and managing tasks within groups
- Setting personal and group goals
- Tracking progress of goals and tasks

8.2. Analysis

Requirement analysis involves refining and organizing the gathered requirements to ensure clarity, consistency, and feasibility. In this phase, duplicate and unclear requirements are removed, and the system is structured into logical modules.

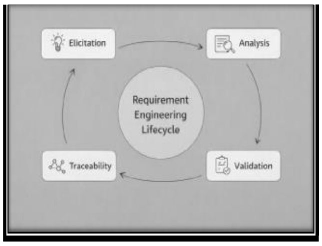
- User Management System (registration, login, profiles)
- Group Collaboration Module (creating/joining groups)
- Task Management System (assignment and tracking of tasks)

8.3. Validation

Requirement validation ensures that the documented requirements accurately represent the needs of users and stakeholders. For this application, validation was done by reviewing each requirement to ensure:

- The system supports real student collaboration scenarios
- All essential features like goal tracking and reputation scoring are included
- No important functionality is missing or duplicated

8.4. Traceability



In this project, traceability is maintained by assigning identifiers to each requirement and ensuring they are connected to:

- Design components
- Implementation features

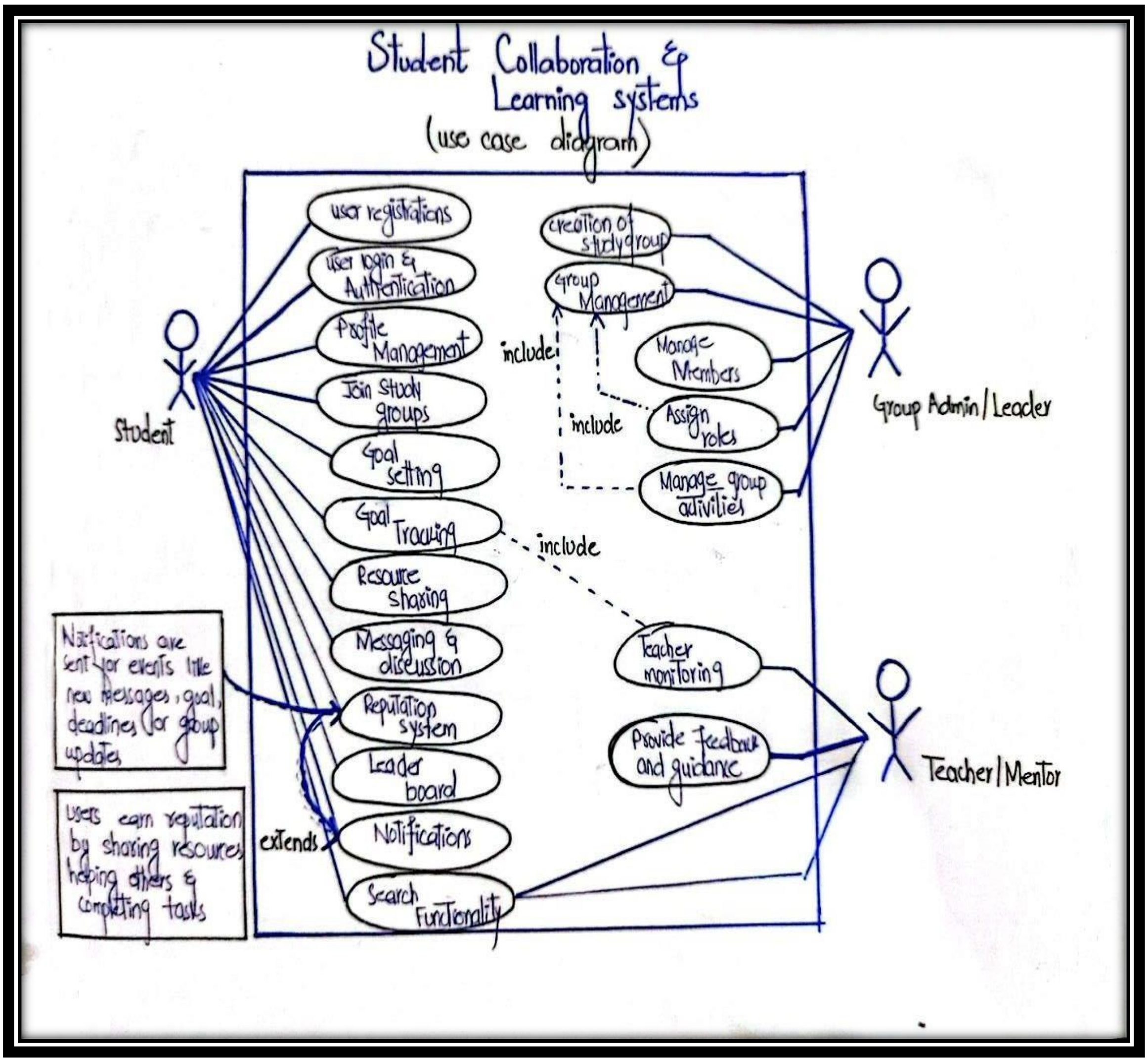
- Test cases (verification of each requirement)

STUDENT LEARNING AND COLLABORATION SYSTEM DIAGRAMS

9.USE CASE DIAGRAM:

A use case diagram is a visual representation of how external users or other systems (actors) interact with a software application to achieve specific goals. As a foundational Unified Modeling Language (UML) tool, it acts as a high-level map of what the system does from the user's perspective

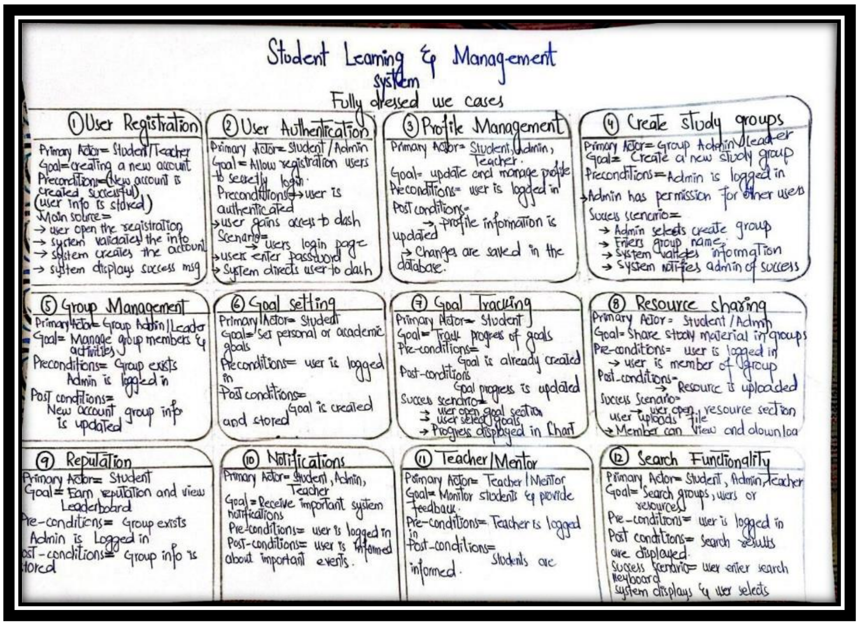
DIAGRAM :



10. FULLY DRESSED USED CASES:

A **fully dressed** use case is a highly detailed, formal document that defines how a user (or external system) interacts with a system to achieve a specific goal. It serves as a comprehensive blueprint for developers and QA testers

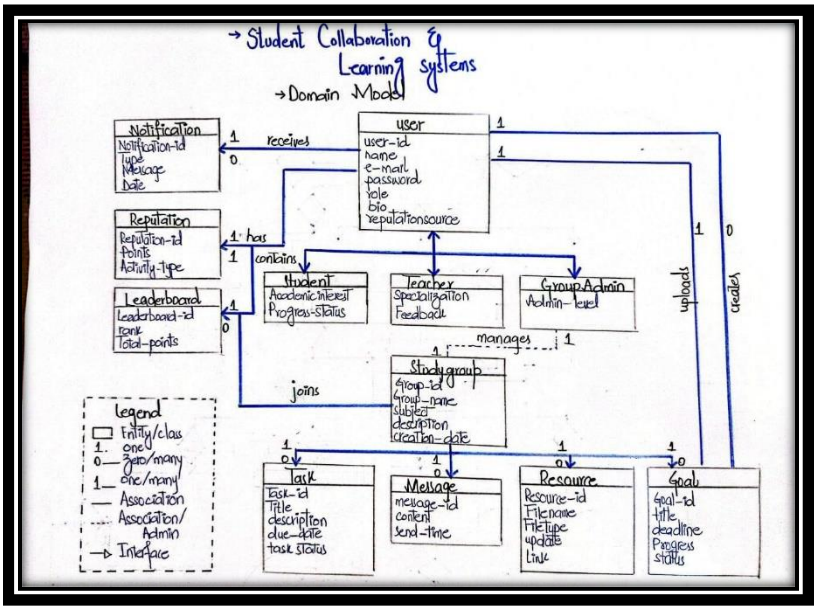
DIAGRAM:



11. DOMAIN MODEL:

A **domain model** is a conceptual representation of the real-world objects, concepts, and rules within a specific subject area (or "domain"). It bridges the gap between business reality and technology, serving as a shared blueprint that defines key terms, data structures, and behaviors

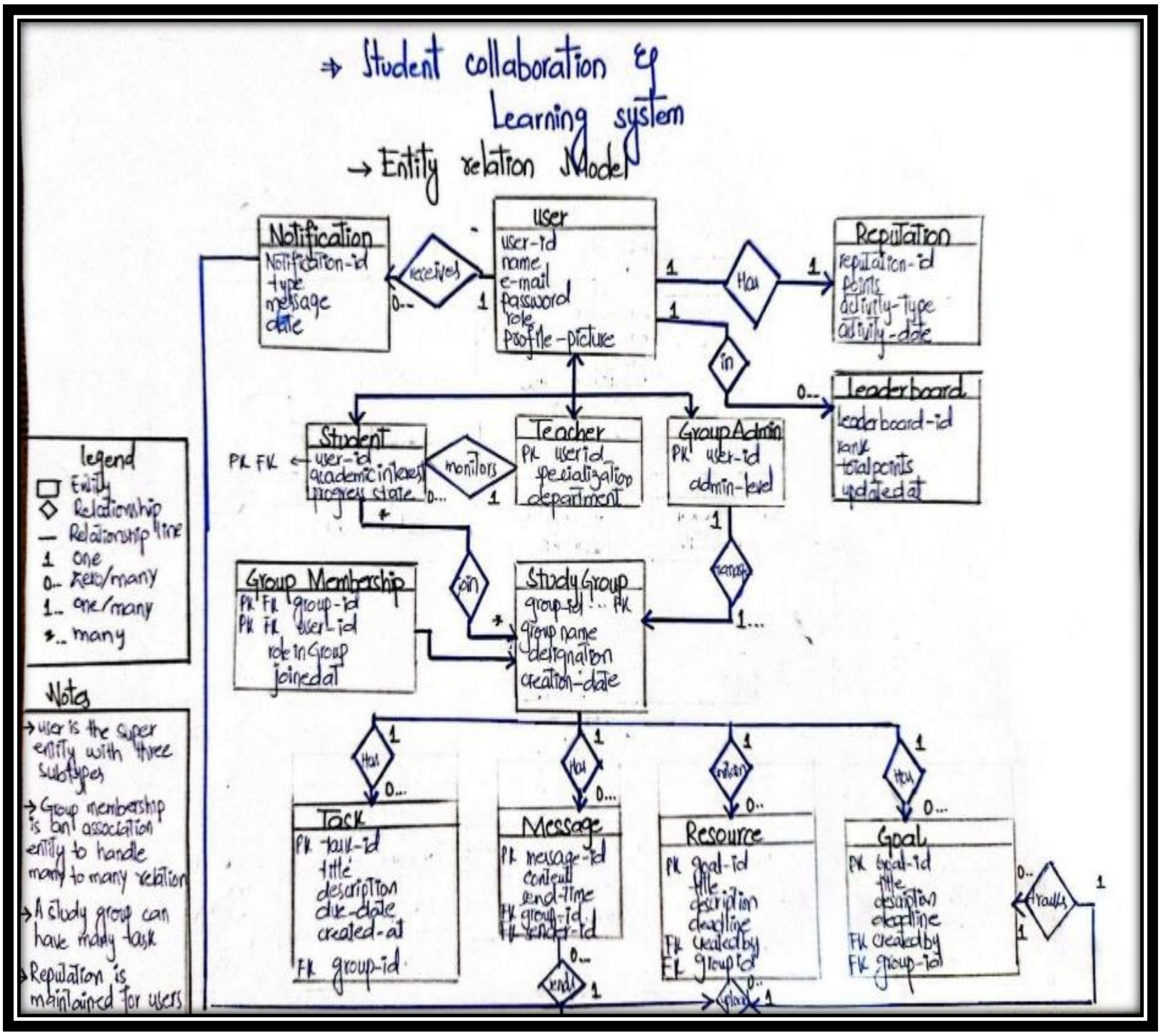
DIAGRAM



12. ENTITY RELATIONSHIP MODEL:

An **Entity-Relationship Model (ERM)** is a conceptual data model used to design databases. It provides a visual and logical map of a system by defining the entities (people, objects, or concepts), their attributes (properties), and how they relate to one another

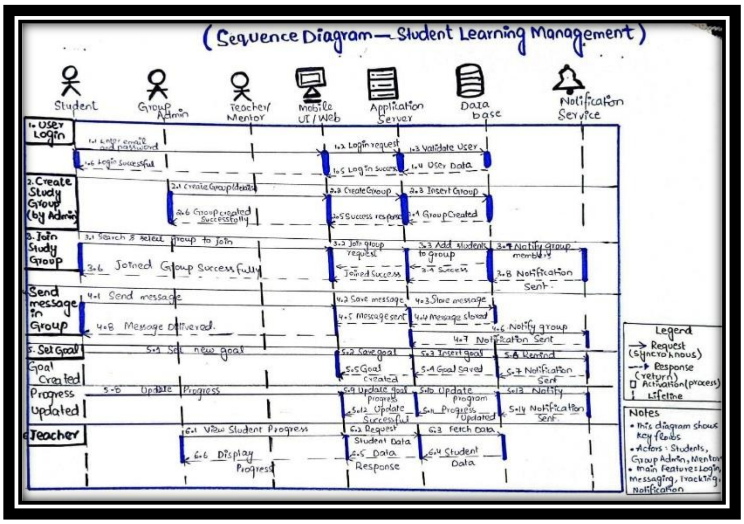
DIAGRAM:



13. SEQUENCE DIAGRAM:

A **sequence diagram** is a visual tool used to show how different parts of a system interact with each other in a sequential order. It maps out the chronological flow of messages, function calls, and data exchanges between objects or components to accomplish a specific task

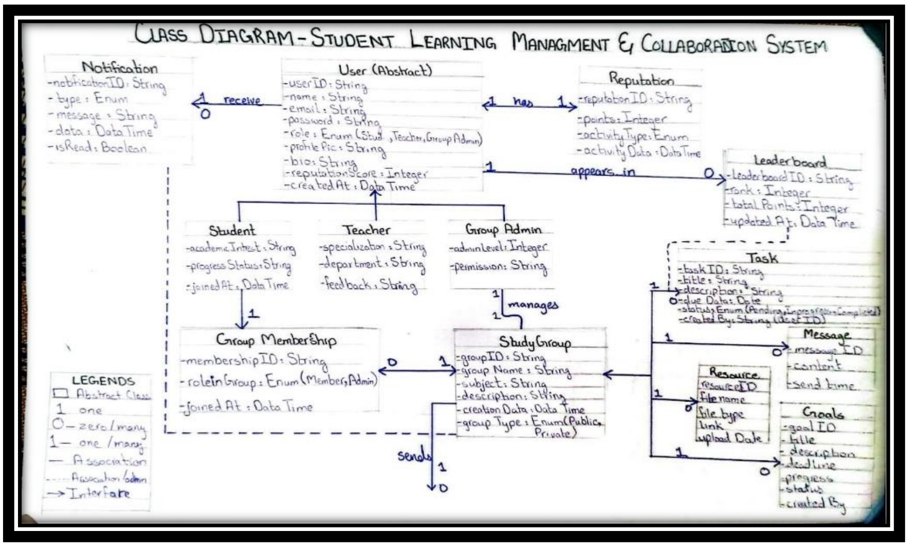
DIAGRAM:



14. CLASS DIGRAM:

A **class diagram** is a structural UML (Unified Modeling Language) diagram that visually represents the static structure of a software system. It acts as a blueprint by mapping out the system's classes, their attributes (data), methods (behaviors), and the relationships between them

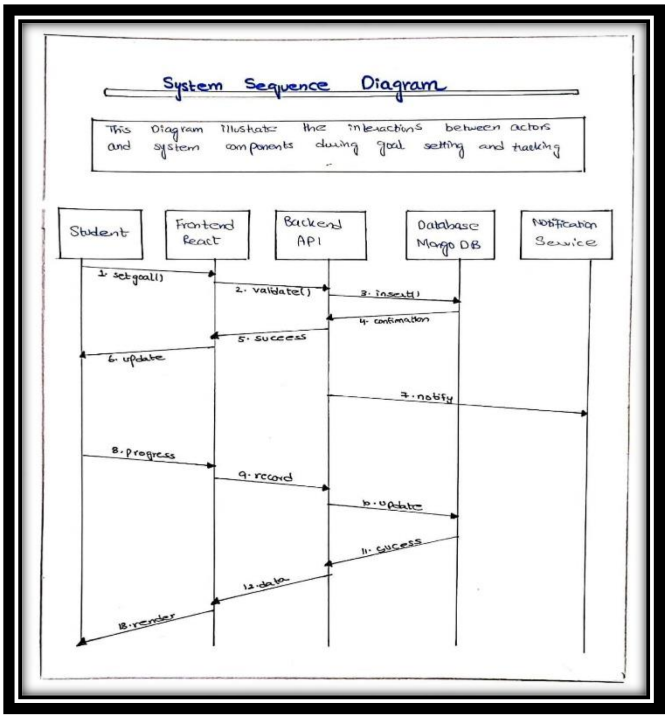
DIAGRAM:



15. SYSTEM SEQUENCE DIAGRAM:

System Sequence Diagram (SSD) is a high-level UML diagram that illustrates the interactions between external actors and a software system during a specific use case. It models the system as a "black box," focusing only on its observable inputs and outputs without showing its internal components or logic

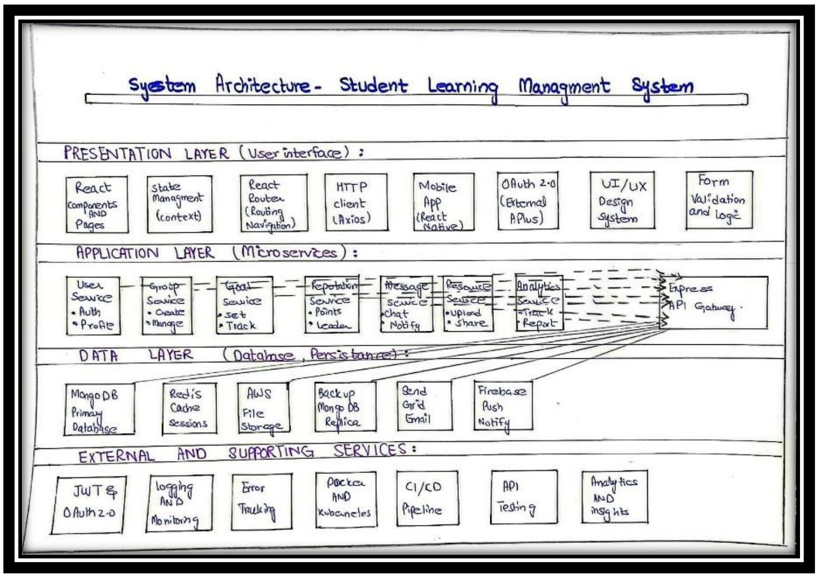
DIAGRAM:



16. SYSTEM ARCHITECTURE

A **system architecture diagram** is a visual blueprint that illustrates the structure, components, and interactions within a hardware or software system. It maps out the "big picture" of a system, acting as a universal map to show how various parts such as user interfaces, databases, servers, and external services communicate and depend on one another

DIAGRAM:



17. Test Cases

The following test cases have been designed to verify that all major functionalities of the Student Learning Management and Collaboration System work according to the specified requirements. These test cases ensure that students, teachers, and group administrators can effectively use the system while maintaining security, reliability, and performance.

17.1 Non-Functional Test Cases

Table 2 non functinal test cases

Test Case ID	Requirement	Test Description	Test Steps	Expected Result	Status
TC-01	FR1 User Registration	Verify that a new user can create an account successfully	1. Open registration page. 2. Enter valid name, email, and password. 3. Click Register	New account is created and confirmation message is displayed	Pass
TC-02	FR1 User Registration	Verify system validation for missing information.	1. Leave required fields empty. 2. Click Register.	Error message appears indicating required fields.	Pass
TC-03	FR2 User Login	Verify login with valid credentials.	1. Enter registered email and password. 2. Click Login.	User is authenticated and redirected to dashboard.	Pass
TC-04	FR2 User Login	Verify login with invalid credentials.	1. Enter incorrect password. 2. Click Login.	Access denied and error message displayed.	Pass
TC-05	FR3 Profile Management	Verify profile update functionality.	1. Open profile settings. 2. Update profile information. 3. Save changes.	Updated information is stored successfully.	Pass
TC-06	FR4 Create Study Group	Verify creation of a study group.	1. Enter group details. 2. Click Create Group.	New study group is created and visible in group list.	Pass
TC-07	FR5 Join Study Group	Verify joining an existing study group.	1. Search group. 2. Click Join Group.	User becomes a member of selected group.	Pass
TC-08	FR6 Group Management	Verify adding members to a group.	1. Select group. 2. Add student.	Student is added successfully.	Pass
TC-09	FR6 Group Management	Verify removing members from a group.	1. Select member. 2. Click Remove.	Member is removed from group.	Pass
TC-10	FR7 Goal Setting	Verify creation of personal academic goals.	1. Enter goal details. 2. Save goal.	Goal is stored and displayed.	Pass

TC-11	FR8 Goal Tracking	Verify progress tracking.	1. Update goal progress.	Progress percentage is updated correctly.	Pass
TC-12	FR9 Resource Sharing	Verify uploading study materials.	1. Upload notes/document. 2. Submit.	File is uploaded and accessible to group members.	Pass
TC-13	FR10 Messaging & Discussion	Verify sending messages in group discussions.	1. Type message. 2. Send.	Message appears in discussion thread.	Pass
TC-14	FR11 Reputation System	Verify reputation score update.	1. Share resource or complete task.	Reputation points increase automatically.	Pass
TC-15	FR12 Leaderboard	Verify leaderboard display.	1. Open leaderboard.	Users are ranked according to reputation scores.	Pass
TC-16	FR13 Notifications	Verify notification generation.	1. Create deadline/task. 2. Wait for notification trigger.	Notification appears successfully.	Pass
TC-17	FR14 Teacher Monitoring	Verify teacher access to student progress.	1. Teacher logs in. 2. Open student report.	Student progress details are displayed.	Pass
TC-18	FR15 Search Functionality	Verify search feature.	1. Search for group/user/resource.	Relevant results are displayed accurately.	Pass

17.2 Performance Testing:

Table 3 performance testing

Test Case ID	Requirement	Description	Expected Result
NTC-01	NFR1 Performance	Load dashboard with normal traffic.	Page loads within 2–3 seconds.
NTC-02	NFR1 Performance	Access multiple resources simultaneously.	No significant delay occurs.

NTC-03	NFR5 Scalability	Simulate large number of users.	System handles increased load efficiently.

17.3 Compatibility Testing:

Table 4 compatibility

Test Case ID	Requirement	Description	Expected Result
NTC-09	NFR8 Compatibility	Open system on Chrome, Firefox, and Edge.	System functions correctly on all browsers.
NTC-10	NFR8 Compatibility	Open system on mobile, tablet, and desktop.	Interface remains responsive and usable.

17.4 Reliability Testing:

Table 5 Reliability

Test Case ID	Requirement	Description	Expected Result
NTC-07	NFR6 Reliability	Continuous system operation for extended period.	No crashes or failures occur.

17.5 Backup and Recovery Testing :

Table 6 Backup And Recovery

Test Case ID	Requirement	Description	Expected Result
NTC-11	NFR9 Backup & Recovery	Simulate accidental data loss.	System restores data successfully from backup.
NTC-12	NFR9 Backup & Recovery	Verify scheduled backups.	Backup process completes without errors.

18. Conclusion:

The **Student Learning Management and Collaboration System** was proposed to address the growing need for a structured academic collaboration platform where students can communicate, learn, share resources, and achieve their educational goals efficiently. Existing communication platforms often contain distractions and lack academic-specific features, making it difficult for students to organize study activities and monitor progress. This project provides a dedicated environment focused entirely on learning and collaboration.

The system incorporates several important functionalities, including user registration and authentication, profile management, study group creation and management, goal setting and tracking, resource sharing, messaging and discussions, reputation scoring, leaderboards, notifications, teacher monitoring, and search capabilities. These features collectively support effective learning, teamwork, accountability, and student engagement.

During the requirement engineering process, user needs were identified, analyzed, validated, and traced to ensure that all essential functionalities were included in the final design. Comprehensive testing was planned through functional and non-functional test cases to verify the correctness and quality of the proposed solutionThe results indicate that the proposed solution satisfies the identified requirements and supports the academic needs of students, teachers, and group administrators.

In conclusion, the Student Learning Management and Collaboration System provides a modern, centralized, and efficient solution for academic collaboration. By encouraging active participation through goal tracking and a reputation mechanism, the system promotes student motivation, improves communication, and enhances learning outcomes.

19. Deployment:

EduCollab: Student Learning Management and Collaboration System

19.1 Introduction

EduCollab is a web-based learning platform developed to support students in managing their academic activities efficiently. The system combines academic tracking, collaboration tools, management, and motivational features within a single environment. It aims to enhance student engagement by promoting active participation, peer learning, and self-improvement.

19.2 Academic Performance Management

One of the core functionalities of EduCollab is its marks tracking module. Students can monitor their academic performance by viewing subject-wise scores, grades, and performance status.

The system automatically calculates grades based on percentages and provides an overall average to help students assess their progress.

Visual indicators such as performance labels allow users to quickly identify strengths and areas requiring improvement. This feature encourages students to remain aware of their academic standing throughout the semester.

19.3 Performance Visualization

To improve understanding of academic results, EduCollab presents performance data through interactive charts. Subject scores are displayed visually, enabling students to compare achievements across courses.

The graphical representation helps users:

- Identify their strongest and weakest subjects.
- Observe overall academic trends.
- Interpret results more effectively than traditional tables

  The charts automatically update whenever new marks are added, ensuring that students always have access to current information.

19.4 Goal Setting and Progress Monitoring

The platform allows students to establish academic goals and monitor their completion over time. Users can define learning objectives and assign target study hours to each goal.

As progress is recorded, the system updates completion percentages and displays visual progress indicators. Completed goals are recognized through achievement notifications and reward points.

This functionality encourages students to:

- Plan their studies effectively.
- Maintain consistency.
- Develop self-discipline and accountability.

19.5 Communication and Collaboration

EduCollab includes a collaborative discussion space where students can exchange ideas, ask questions, and support one another academically.

Through the chat system, users can:

- Participate in study discussions.
- Seek clarification regarding difficult topics.
- Share suggestions and solutions.
- Receive encouraging responses from the mentor component.

The collaboration environment promotes peer learning and creates a sense of community among students.

19.6 Feedback Management

The feedback module provides students with an opportunity to express opinions regarding courses and platform services. Users can submit suggestions, concerns, or appreciation messages.

Submitted feedback becomes visible within the system, encouraging transparency and continuous improvement. This feature enables educators and administrators to understand student perspectives and make informed decisions.

19.7 Reputation and Leaderboard System

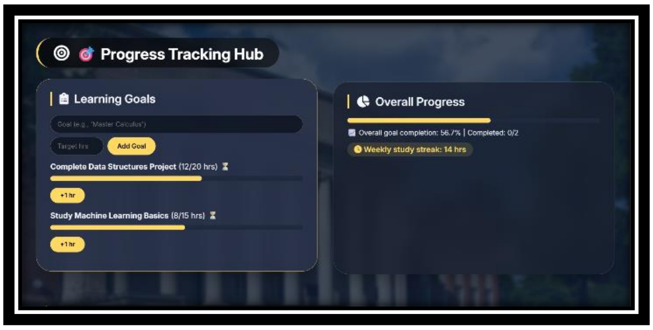
To enhance motivation, EduCollab incorporates gamification techniques through a reputation system. Students earn points by participating in various activities such as:

- Sending messages.
- Sharing learning resources.
- Completing goals.
- Providing feedback.
- Contributing to collaborative activities.

Accumulated points determine rankings on the leaderboard, encouraging healthy competition and active engagement. Achievement badges further recognize consistent effort and participation.

19.8 Learning Resource Sharing

The platform serves as a repository for educational materials where users can share useful learning resources.



Students can contribute:

- Educational websites.
- Tutorial links.
- Documentation references.
- Course materials.

This feature promotes collective knowledge building and provides easy access to valuable study content for all participants.

19.9 Notification Services

EduCollab keeps users informed through a real-time notification system. Important updates such as goal achievements, new resources, feedback submissions, and academic activities are communicated instantly.

19.10 Search Functionality

The global search feature allows users to locate information quickly across different sections of the system.

Students can search for:

- Study groups.
- Shared resources.
- Personal goals.

This functionality improves navigation efficiency and enhances user experience by reducing the time required to access relevant information.

19.11 User Profile Management

Each user maintains a profile containing essential academic and personal details. Profile information may include the student's name, interests, biography, and accumulated reputation score.Users can personalize their profiles, enabling better identification and fostering stronger interaction among members of the learning community.

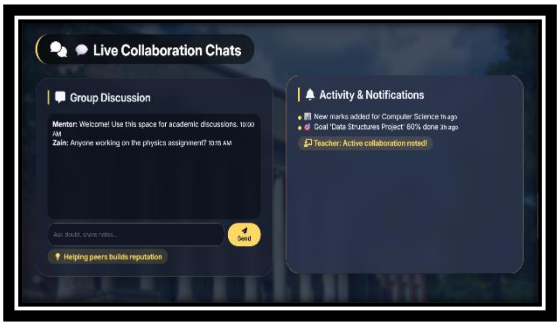
19.12 Teacher and Mentor Monitoring

EduCollab also supports educators by providing an overview of student engagement and performance.

Teachers can monitor:

- Student participation levels.
- Reputation scores.
- Goal completion status.
- Collaborative involvement.

This information helps instructors identify students requiring additional support and recognize highly active participants.



19.13 User Interface and Experience

The system adopts a modern and visually appealing interface designed to maximize usability. Responsive layouts ensure accessibility across multiple devices, including desktops, tablets, and smartphones. Interactive design elements contribute to an engaging learning environment while maintaining clarity and ease of navigation.

19.14 Data Handling and Storage

The current implementation operates using session-based data storage. Information remains available during the active session and resets upon page refresh. However, the architecture can be extended to support cloud databases and backend services in future implementations.

19.15 Cross-Platform Compatibility

EduCollab is designed to function effectively on different screen sizes and modern web browsers. Responsive design principles ensure consistent performance regardless of the device being used.

19.16 Intended Users and Applications

The platform is suitable for several educational contexts, including:

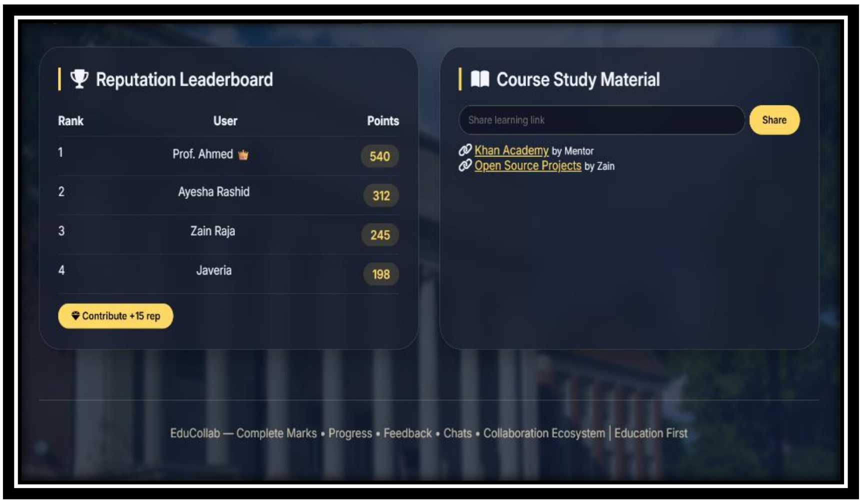
- Individual students seeking to monitor academic progress.
- Study groups collaborating on assignments and projects.
- Teachers supervising student engagement.

19.17 Technical Overview

The system has been developed using standard web technologies, including HTML, CSS, and JavaScript. Additional libraries are utilized for data visualization and interface enhancement. A major advantage of the project is that it operates without requiring a dedicated backend server or database during demonstration, making it easy to deploy and present.

20. Conclusion

EduCollab provides an integrated solution for academic management, collaboration, and student motivation. By combining performance tracking, communication tools, goal management, resource sharing, and gamification features, the platform creates a supportive digital learning environment. The system demonstrates how technology can improve educational experiences by promoting organization, engagement, and continuous growth among learner.



References

Get the idea and information related to some references mentioned.

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Submission Details

File Name: Assignment No. 04, Student Learning Management and Collaboration System

Submission Date: 16th June 2026

Word Count: 6,400 – 6,600 words

Character Count: 43,000 – 45,000 characters